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An Integrated Framework for the Assessment of Automated Driving Systems

Martin Skoglund

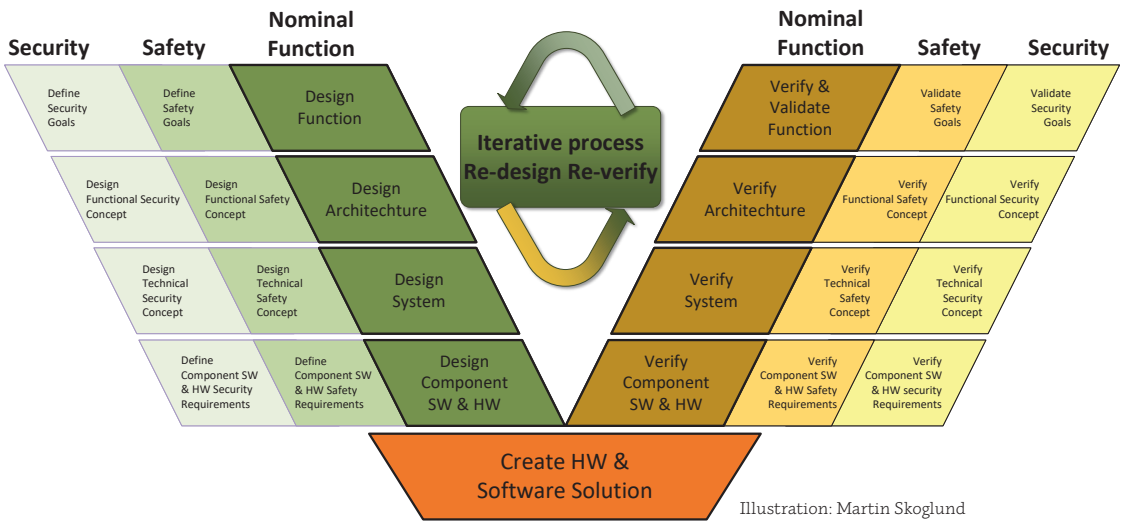


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AN INTEGRATED FRAMEWORK FOR THE ASSESSMENT OF AUTOMATED DRIVING SYSTEMS

Martin Skoglund

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Department of Computer Science & Engineering

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Martin Skoglund

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Institutionen för datavetenskap och datateknik

Abstract

The road transport system is undergoing major transformations, notably the transition to fossil-free propulsion and advances in automation.

In the automotive domain, highly automated vehicles are typically referred to as incorporating Automated Driving Systems (ADS). Despite significant technological progress and anticipated benefits, the widespread deployment of these systems remains limited. This slow rollout is largely due to the increased complexity at higher automation levels, where vehicles must take on the full driving task and transition to fail-operational architectures capable of managing faults without human intervention.

The increasing complexity of these systems stems from a growing reliance on new, often noisy sensors and external connectivity for perception and decision-making. In addition to established safety concerns, cybersecurity has become essential, as communication interfaces and complex sensors introduce new vulnerabilities. Advances in environmental sensing further require attention to functional limitations that arise independently of malicious interference. Taken together, these developments present challenges in forming a comprehensive understanding of the contexts in which the systems are intended to operate safely. This, in turn, hinders the development of effective validation procedures and complicates efforts to substantiate claims of safety, security, and overall trustworthiness.

This PhD thesis advances the assurance of ADSs by integrating safety and cybersecurity into a unified assessment framework. The contributions are threefold: (1) context-dependent requirements; introducing technology-specific assessment templates for positioning, communication, and cybersecurity to enable more effective evaluation, together with a formalized operational design domain (ODD) for improved coverage and traceability; (2) multi-concern development management; demonstrating that co-engineered safety and security processes enhance completeness, supported by structured argument patterns, unified process models, and synchronization mechanisms to align assurance activities; and (3) independent assessment methods; developing a complementary testing approach called Assessment of Cybersecurity-informed Safety (AoCiS) and a systematic method for allocating test cases (METAFODD) to appropriate environments, enabling scalable scenario-based validation. Case studies and demonstrations show gains in efficiency, clarity, and completeness, while large-scale validation remains a challenge to be addressed in the future. Overall, the research presented in this thesis establishes structured and transparent methods to support independent and systematic assurance of safety and security in future evaluation efforts.

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Abstract

The transport system is in a period of profound transformation, driven by the shift to fossil-free propulsion and the gradual introduction of automation. The anticipated introduction of connected, cooperative, and automated mobility will have far-reaching implications for society. Despite notable technological progress and the societal benefits often highlighted, the broad implementation of automation has yet to materialize.

The limited rollout can largely be attributed to the significant complexity associated with higher levels of automation, where the vehicle must assume the entire driving task and maintain functionality even under fault conditions. The growing complexity of these systems stems from an increased reliance on sophisticated sensors and connectivity to support perception and decision-making. Beyond established safety concerns, cybersecurity has become critical, as communication interfaces and advanced sensing technologies introduce new vulnerabilities. Furthermore, progress in environmental sensing demands careful consideration of inherent functional limitations, even in the absence of malicious interference.

High interdependence comes with challenges in establishing a clear understanding of the contexts in which the systems are intended to operate safely. The complexity slows the development of effective validation procedures and complicates efforts to substantiate claims of safety, security, and trustworthiness.

This PhD thesis advances assurance of advanced driving systems (ADS) by integrating safety and cybersecurity in a unified framework. The contributions are threefold: (1) *context-dependent requirements*; introducing technology-specific assessment templates for positioning, communication, and cybersecurity to enable more effective evaluation, together with a formalized operational design domain (ODD) for improved coverage and traceability; (2) *multi-concern development management*; demonstrating that co-engineered safety and security processes enhance completeness, supported by structured argument patterns, unified process models, and synchronization mechanisms to align assurance activities; and (3) *independent assessment methods*; developing a complementary testing approach called Assessment of Cybersecurity-informed Safety (AoCiS) and a systematic method for allocating test cases (METAFODD) to appropriate environments, enabling scalable scenario-based validation. Case studies and demonstrations show gains in efficiency, clarity, and completeness, while large-scale validation remains a challenge to be addressed in the future. Overall, the research presented in this thesis establishes structured and transparent methods to support independent and systematic assurance of safety and security in future evaluation efforts.

Sammanfattning

Transportsystemet befinner sig i en period med omfattande förändring, präglad av övergången till fossilfria drivmedel och en gradvis introduktion av automatiserade funktioner. Den förväntade introduktionen av uppkopplad, kooperativ och automatiserad mobilitet bedöms få genomgripande konsekvenser för sektorn som helhet. Trots betydande tekniska framsteg och den betydance samhällsnytta som lyfts fram som fördelar har en bred implementering av automatisering ännu inte realiserats. Den begränsade utrullningen kan i huvudsak förklaras av den betydande komplexitet som följer med högre automatiseringsnivåer, där fordonet måste kunna överta hela köruppgiften och upprätthålla funktion även under felförhållanden.

Vid högre automatiseringsnivåer krävs att systemet inte enbart ersätter förarens kontinuerliga övervakning, utan även självständigt upptäcker, analyserarar och hanterar fel. Högre automatiseringsnivåer förlitar sig på avancerade varseblivningsalgoritmer samt ett ökat beroende av kommunikation, positionering och cybersäkerhet för att möjliggöra ett robust och välgrundat beslutsfattande. Dessa beroenden medför nya utmaningar, bland annat ökad systemkomplexitet och svårigheter att entydigt avgränsa systemet. Detta försvårar i sin tur fastställandet av tillförlitliga testprocedurer för funktionssäkerhet, cybersäkerhet och systemprestanda.

Denna doktorsavhandling behandlar en förbättrad insyning av automatiserade körsystem (ADS) genom att integrera funktionssäkerhet och cybersäkerhet i ett sammanhållet insyningsramverk. Arbetet har gett tre huvudsakliga bidrag. Det första avser krav, dikterade av kontexten, vilket omfattar mallar för teknikspecifika bedömningar av positionering, kommunikation och cybersäkerhet samt en formaliserad operativ designdomän (ODD) som underlättar strukturerad miljöparameteranalys och spårbarhet. Det andra bidraget rör parallell utveckling, där samordnade processer för funktionssäkerhet och cybersäkerhet stärker fullständigheten, vilket stöds av argumentmönster, processmodeller och synkroniseringsmekanismer. Det tredje bidraget gäller metoder för oberoende utvärdering: scenariosbaserad säkerhetsanalys (AoCiS) och en metod för allokering av testfall (METAFODD) till ändamålsenliga testmiljöer som möjliggör en skalbar scenariosbaserad validering. Fallstudier påvisar en förbättrad tydlighet och struktur med hälp de förslagna kompletteringarna av insyningsramverken, medan frågor kopplade till storskalig validering kvarstår som ett fortsatt forskningsbehov.

When technology transcends what we can define,
it echoes arcane design.
So ergo, hence, do you agree,
what we try to assess, is wizardry?

Acknowledgments

Firstly, I would like to express my sincere gratitude to my supervisors, Professor Hans Hansson and Professor Sasikumar Punnekkat at Mälardalen University, for their enduring support, insightful guidance, and trust in my work throughout this journey. Their measured balance of direction and independence has been central to the progress and completion of this research.

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Martin Skoglund
Borås, 2025

List of Publications

Papers included in thesis[1]

Paper A: Martin Skoglund, Fredrik Warg, Anders Thorsén and Mats Bergman. Enhancing Safety Assessment of Automated Driving Systems with Key Enabling Technology Assessment Templates In *Vehicles*. 2023. MDPI. [1].

Paper B: Martin Skoglund, Fredrik Warg, Behrooz Sangchoolie. In search of synergies in a multi-concern development lifecycle: Safety and cybersecurity. In *International Conference on Computer Safety, Reliability, and Security* . SAFECOMP Workshops, 2018. [2].

Paper C: Fredrik Warg, Martin Skoglund. Argument Patterns for Multi-Concern Assurance of Connected Automated Driving Systems. In *4th International Workshop on Security and Dependability of Critical Embedded Real-Time Systems*. CERTS, 2018. [3].

Paper D: Martin Skoglund, Fredrik Warg, Hans Hansson, Sasikumar Punnekkat. Synchronization of an Automotive Multi-Concern Development Process. In *40th International Conference on Computer Safety Reliability and Security*. SAFECOMP Workshops 2021. [4].

Paper E: Martin Skoglund, Fredrik Warg, Hans Hansson, Sasikumar Punnekkat. Black-Box Testing for Security-Informed Safety of Automated Driving Systems. In *Proceedings of 93rd Vehicular Technology Conference*. VTC spring 2021. [5]

Paper F: Martin Skoglund, Fredrik Warg, Hans Hansson, Sasikumar Punnekkat. Formalizing Operational Design Domains with the Pkl Language. In *Proceedings of the 36th Intelligent Vehicle Symposium*. IV 2025. [6]

Paper G: Martin Skoglund, Fredrik Warg, Hans Hansson, Sasikumar Punnekkat. Methodology for Test Case Allocation based on a Formalized ODD. *Accepted to the 20th International Workshop on Dependable Smart Embedded Cyber-Physical Systems and Systems-of-Systems*. DECSoS 2025. [7]

[1]The included papers have been reformatted to comply with the thesis layout.

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Part I

Thesis

Chapter 1

Introduction

Connected, Cooperative, and Automated Mobility (CCAM) promises substantial societal benefits, as stakeholders in the CCAM community [8] anticipate improvements in road safety through the reduction of human-induced incidents, environmental gains through decreased emissions and congestion, and enhanced inclusiveness in terms of mobility and access to goods. At the same time, the road transport system is already undergoing profound changes, including the transition to fossil-free propulsion and the gradual adoption of higher levels of automation. CCAM, as an overarching concept relevant for the entire transport ecosystem, is therefore closely linked to these developments and is expected to contribute significantly to key United Nations Sustainable Development Goals [9]. The scope of CCAM is immense, encompassing all forms of transport, be it by land, sea, or air. Many significant issues in the certification process are similar across all transport modes, e.g., presenting a structured argument supported by evidence that the system is safe. The automotive domain can serve as a representative example. The work presented in this thesis exclusively focuses on land transport and the advancements being made for road vehicles, especially in ADS. In the automotive domain, highly automated vehicles are commonly referred to as Automated Driving Systems (ADS) [10]. Despite the positive expectations and significant investments, the widespread deployment of such systems remains constrained, with availability currently limited to pilot programs. This limited adoption reflects the fact that many societal, regulatory, technical, economic, and operational challenges remain unresolved, and that the broader implications of CCAM technologies for safety, cybersecurity, and the environment are not yet sufficiently understood. Consequently, users and policymakers remain cautious, in contrast to the technological maturity that manufacturers claim would already support large-scale deployment.

A means forward is to build trust by providing evidence that the system is safe and secure through a harmonized and transparent validation procedure. Connected and cooperative solutions can be seen as enablers for the mobility automation capabilities in CCAM. Thus, the validation procedure of CCAM focuses on ensuring that the automation component of the implementation aligns with the stakeholders' requirements. The principal challenge for future safety and security validation lies in the wide variety

of driving situations that must be systematically tested and evaluated for the automated function under consideration. One approach is to represent the operational parameters of the traffic environment as a series of temporal instances, where a sequence of such instances constitutes a scenario. Scenarios serve as the fundamental units for evaluating behavioral safety, and scenario-based testing has therefore gained increasing prominence. Proving top-level safety claims through scenario-based testing is independent of the underlying implementation technology and therefore remains agnostic to AI; the articulation of AI-related aspects belongs in other parts of safety assessment frameworks for ADS, beyond the scope of direct performance evaluation, and remains an open consideration. This development is supported by the ISO 3450x series of standards, which define test scenarios for automated driving systems [11, 12, 13, 14, 15]. It can be complemented by established verification frameworks, including ISO 29119 on software and systems engineering and software testing [16].

The scope of scenario-based testing depends on the complexity of the intended automation task to be solved and the environment in which it is intended to operate. Currently, available procedures do not provide efficient and cost-effective solutions for this new level of automation, even though the community has been very active recently, discussing, debating, and evolving proposals concerning the future of validation of automated vehicles and making significant contributions to expanding the frontiers of consensus on how to tackle the problem. The complexity of the issues is of such dignity that the progression to a practical solution is hard to predict. The uncertainty in implementing the validation procedure, particularly regarding how certification authorities are expected to utilize it, remains.

This increased automation requires that the chief design principle no longer be that of a fail-safe system, i.e., systems that can become unavailable when a problem occurs. ADS with level 3 and above features, according to SAE J3016:2021 [10], aim to remove active supervision by a human and replace the driver with fail-operational ADSs, i.e., an automated function for which the driver is not the fallback.

The operational domain in which a vehicle is intended to function represents a foundational aspect of both the development and assurance of automated driving. In contrast to conventional vehicles equipped with advanced driver assistance systems (ADAS), ADS [10] are closely interwoven with safety-critical vehicle functions. To provide the necessary contextual basis, the operational design domain (ODD) [13] specifies the external conditions that influence the driving task, thereby delineating the scope of operation and the coverage criteria for testing within the ODD. Demonstrating that the fail-operational driving task is thoroughly verified and validated across the entire ODD is therefore essential. This task is considerably more demanding than, for instance, demonstrating that an ADAS avoids excessive false positive activations, which primarily risks reducing user trust and the perceived premium value of the function, rather than compromising overall safety.

The state space spanning the combination of operational conditions to be verified and validated in an ADS is vast, making it infeasible to achieve adequate coverage with a finite number of physical tests. Developers of ADS and independent testing cannot drive [17] their way to safety, even less so when considering security simultaneously. Any building block or complete method for future ADS certification must supplement

real-world testing with accelerated testing, applicable under a type approval or self-certification regime. One approach is to establish testing and validation procedures for ADS functionalities that systematically integrate simulation-based evaluations with tests conducted in physical environments. Conventional functions that have traditionally been examined exclusively through physical test suites and classical methods could likewise benefit from a rigorous and transparent methodology that combines both simulated and real-world evaluations.

The Validation Methods for Automated Driving (VMAD) is an informal working group operating under the Working Party on Automated/Autonomous and Connected Vehicles (GRVA), which is directed by the United Nations Economic Commission for Europe (UNECE). This group has proposed a process framework intended to serve as the basis for future certification procedures in Europe. The proposed certification is called the New Assessment/Test Method for Automated Driving (NATM) [18] and is based on a multi-pillar approach. In NATM, accelerated testing, coupled with validity documentation in the audit and assessment pillar, complements, rather than substitutes, classical test track certification. Proving ground tests, such as classical certification elements, are essential for the suggested multi-pillar approach. The process is geared toward SAE automation levels 3 through 5. It is recommended to rely on a combination of pillars to validate the capabilities of an ADS that cannot be comprehensively or effectively accomplished through a single validation methodology alone.

The research presented in this thesis aims to establish trust in ADS by contributing to a harmonized and transparent validation procedure. The proposed procedure assesses the interplay between safety and security to determine the appropriateness of the intended use of the ADS and confirms the adequacy of risk reduction measures.

The operational context and the key enabling technologies employed in ADS engineering necessitate the management of a massive and complex set of requirements. A verdict on the appropriateness of intended use can be reached only through a clear understanding of the intricate requirements imposed on the vehicle. Therefore, an evaluation of the safety and security requirements applicable to ADS was conducted, which demonstrated the need to address these concerns simultaneously. As a result, a proposal for an argumentation pattern that jointly covers safety and security requirements, including their overlaps, was developed. The complexity of these requirements can thereby be managed through a traceable argument structure, supported by a documented rationale that preserves the reasoning underlying the design.

Further analysis examined potential synergies between functional safety and security standards. The study revealed synergies in the interaction between safety and security throughout the development process of an ADS. The extension of this analysis into a unified process model provides evidence that the safety and security goals were achieved. Within this model, the argumentation patterns for safety and security are combined and systematically mapped to a process for demonstrating conformance with relevant standards.

The overlap in assumptions made in the operational context extends into their engineering translation into requirements and the implementation of risk mitigation measures. If left unaddressed, inconsistencies between safety and security can propagate across the development process. This risk can be minimized by the proposed

synchronization mechanism for safety and security interaction, which ensures a structured alignment of concerns.

The contributions described in this thesis serve as prerequisites for introducing a novel building block in a future validation process. Specifically, the proposed subprocess, termed Assessment of Cybersecurity-informed Safety (AoCiS), supplements real-world testing with co-simulation and is designed to enable independent assessment of security-informed safety while reducing the risks inherent in physical testing. This sub-process is designed to be integrated into the multi-pillar approach, establishing a comprehensive evaluation framework for ADS.

1.1 Thesis outline

The thesis is organized into two major parts. Part I of the thesis aims to contextualize the research, and Part II presents the contributions, comprising seven peer-reviewed articles that detail the work.

The remainder of Part I is organized as follows:
Chapter 2 Background and related work provides an understanding of the terminology and the technical context to support the understanding of the challenges and the motivation to address them.
Chapter 3 Research summary covers the methodologies used within the conducted research and the specific motivation of the detailed research questions.
Chapter 4 Contributions summarizes the contributions of the included articles and progress towards answering the formulated research questions.
Chapter 5 Summary and conclusion summarizes the thesis, along with ideas for future investigations.

Chapter 2

Background and related work

This section presents background and related work to provide a context for the work in Part II of the thesis.

An automated driving system is the hardware and software that realizes the automated driving functionality in vehicles, sometimes referred to as autonomous vehicles or self-driving vehicles. Higher levels of automated driving functions enable the transfer of control tasks from the driver to the vehicle system, even in complex traffic environments. However, in this context, autonomy or self-driving can be misinterpreted as the capacity for self-governance, which is a misnomer for the systems under consideration. Hence, we use the term automated.

The technology under consideration is framed using the widely adopted SAE J3016 taxonomy [10], which offers a standardized set of terms and definitions intended to support a consistent understanding of driving automation. Earlier foundational work, such as the PATH program's articulation of automated highway system concepts [19], as well as subsequent efforts by BASt [20], NHTSA [21], and Yang et al. [22], have contributed to the development of classification schemes for automated driving. These works present alternative formulations and conceptualizations of automation levels, although they do not constitute systematic comparative analyses of taxonomies in the strict sense. For an ADS [10], features (i.e., SAE automation Levels 3 to 5) are capable of performing the complete dynamic driving task (DDT), crash mitigation, and avoidance. ADSs aim beyond fail-safe behavior and intend to remove active supervision by a human driver, replacing it with fail-operational systems. An operational design domain (ODD) defines the operating conditions within which an ADS is designed to function. Consequently, such systems must have fall-back strategies that can always ensure continued operation or take the system to a state where the risk is acceptably low in case of failures of in-vehicle components or if the vehicle is approaching the boundaries of the ODD. However, in addition to design challenges, the complexity of the fail-operational task requires new and more effective methodologies for verification and validation.

An intricate dependence on the entire mobility infrastructure characterizes efficient solutions for ADS [23, 24]. The interdependency between different aspects relies on

novel key technologies, such as Artificial Intelligence, Big Data, and cybersecurity, to enable effective solutions. Vehicle technologies that integrate into the transport system infrastructure foster the need to validate the system. There is also a need to handle the perceived conflict between technological innovation and existing regulations [25]. Artificial intelligence and big data pose significant challenges for ethically aligned design and privacy [26, 27], in addition to the direct safety-impact of using AI in an ADS, which remains largely unaddressed. An important aspect in the design and development of an automated driving system is the careful configuration of its sensing and communication architecture, ensuring that computational and physical resources are utilized appropriately and the system remains resilient against single-point failures and cybersecurity threats [28].

2.1 Requirements on ADS

The first step in addressing the specific challenges of developing an ADS for the European market, particularly concerning type approval and prototype testing exemptions, is to identify the relevant requirements and constraints. Analysis indicates that safety, security, and human–system interaction are fundamental considerations throughout ADS development. Foundational guidance is provided by the Functional Requirements for Automated Vehicles (FRAV) initiative under the UNECE's Working Party on Automated/Autonomous and Connected Vehicles (GRVA) [29]. Additionally, the HEADSTART project defines testing and validation procedures for the functions of automated driving systems. Insights from HEADSTART [30] highlight a largely unmet need for verification and validation of key enabling technologies, serving as a practical specification of current challenges. Complementing these findings, Sobiech et al. [31] emphasize that assurance frameworks must integrate technical safety arguments with policy-level requirements, ensuring regulatory and societal expectations are explicitly addressed. The collective data from European stakeholders thus provides a valuable overview of prerequisites for both development and testing.

Building upon these foundations, the SUNRISE project [32] developed and demonstrated a harmonized and scalable Safety Assurance Framework (SAF) for Cooperative, Connected, and Automated Mobility (CCAM) systems. It addressed the need for a common European validation framework by integrating scenario-based testing methodologies, combining virtual and physical testing, and fostering collaboration among stakeholders. The project focused on establishing a federated European scenario database framework [33] to pave the way for developing comprehensive testing tools and integrating functional safety and cybersecurity considerations into the framework. They constituted a substantial contribution toward enabling the large-scale deployment of automated vehicles on European roads.

The relationship between this thesis and SUNRISE is reciprocal. Results from the thesis, such as the formalization of operational design domains and cybersecurity-informed safety methods, have provided input relevant to ongoing discussions in the project. At the same time, SUNRISE has offered practical perspectives and feedback that have informed the development of the thesis work, particularly in aligning methodological contributions with broader European efforts toward harmonized validation.

This exchange of knowledge has supported progress in both academic research and the project context.

The standardization of ODD definitions is fundamental to the safety of automated driving systems. Established standards such as ISO 34503 [13] and BSI PAS 1883 [34], together with recent initiatives including ASAM OpenODD [35] and OpenTestSpecification [36], provide structured methods for specifying operational conditions. The completed ASAM OpenODD project has introduced a machine-interpretable, technology-independent conceptual model for ODD representation, offered formal structures for describing operational, current, and target domains, and enabled consistent exchange of ODD information across tools and testing environments. This supports a more unified basis for scenario generation, simulation workflows, and validation activities. Despite these advances, limited guidance is available on how ODD definition processes should be integrated with end-to-end test workflows or real-time monitoring functions [37, 38]. Further contributions, such as the framework proposed by Gyllenhammar et al. [39], outline how ODD descriptions may inform safety argumentation, indicating a growing recognition of the ODD as a central artefact in assurance activities.

Functional safety in road vehicles is governed by the automotive-specific international standard ISO 26262, which was introduced in 2011 and revised in 2018 [40]. ISO 26262 is based on the general functional safety standard IEC 61508 [41] but tailored to the particular needs and context of the automotive sector.

Safety of the Intended Functionality ISO/PAS 21448:2019 applies to functionality in an automotive component that requires the absence of unreasonable risk due to hazards resulting from functional insufficiencies of the intended functionality or by reasonably foreseeable misuse to be considered adequately safe [42]. In other words, ISO/PAS 21448:2019 ensures the safety of the intended functionality (SOTIF) without any faults.

In addition to traditional safety issues and sensor performance, there is also a high degree of dependence on cybersecurity to achieve this new level of automation, such as the use of external communication and environmental sensors that are susceptible to remote attacks. Analyzing the threats on automated vehicles shows the need for high redundancy in implementing vehicle automation systems [43]. This increased reliance on connectivity and sensors makes the cybersecurity standard ISO 21434 [44] and regulation 155 [45] for cybersecurity for road vehicles highly relevant.

The applicable standards can be narrowed down to a non-dispensable minimum relevant to the specific implementation and domain. The appropriate standards for a non-dispensable minimum are ISO 26262 [40] (addressing functional safety), ISO/PAS 21488 [42] (addressing the safety of the intended functionality, SOTIF), and ISO/SAE 21434 [44] (addressing cybersecurity). The technical report ISO 5083 [46] defines a high level harmonized framework for multi-concern development. However, the practical implementation and full impact of this technical report remain to be seen; it may evolve into a recognized standard if the ongoing work matures and achieves wider consensus. In the meantime, there remains a pressing need for practical guidelines and best practices for multi-concern development, building on existing and emerging standards. Furthermore, robust methods are necessary to assess conformity with a recognized minimum set of standards, ensuring the dependability and security of

systems.

Safety for road vehicles is typically demonstrated by conformance to the ISO 26262 standard through a safety case, which consists of artifacts that document the mapping of safety activities against normative requirements. Explicit arguments demonstrating how the product-specific safety rationale aligns with the overall safety framework of the standards provide structure, stringency, and clarity to the development process. Any deviation from the framework or tailoring thereof in the development process must be documented with a rationale. Burden et al. [25] show that structured stakeholder collaboration, such as through policy labs, can reduce regulatory uncertainty and strengthen the alignment of safety cases with approval processes. When the interaction between concerns is planned and limited, as proposed by Warg and Skoglund [3], there is an excellent possibility of retaining the benefits of co-engineering without the extra effort of high-frequency interaction between different disciplines, such as safety and security [47]. The premise is that such an argument is even more critical - even if the standards may not mandate it - when the complexity increases, for instance, when demonstrating that all potential security risks that could compromise the system's safety have been addressed. By assessing the requirements of the development process and the product of an ADS, concurrent conformance to several standards is needed.

2.2 Multi-concern assurance

Standards conformance is intended to ensure that quality concerns are addressed appropriately. There is, however, a lack of experience in dealing with security in safety engineering and vice versa. Even though fundamental requirements for collaboration between concerns exist, e.g., ISO 26262 [40] expresses the need to consider interconnection with cybersecurity, there is little or no guidance on transforming these requirements into a practical process that enables conformance to several standards with different concerns. Prior work has shown that while safety and security assurance are often treated separately, their interaction is unavoidable and can only be effectively managed through integrated approaches [48].

A promising approach to the assurance and certification of safety-critical systems is to adopt a holistic, reuse-oriented strategy based on architecture-driven assurance, referred to as multi-concern assurance. In this context, the main elements of the AMASS approach [49] can be applied to model aggregated conformance requirements across multiple standards.

During the design phase, one central objective is to eliminate inconsistencies in safety and security requirements and establish a shared understanding of the assumptions embedded in the operational context [4]. As development progresses through implementation and testing, and later after market release, the interaction between safety and security can be described as a dynamic interplay between realized functions. Human involvement introduces further complexity, as operational assumptions and system usage must also be validated against realistic human–system interaction scenarios [50]. Taken together, the intricate co-dependence of safety and security underscores that they are most effectively addressed in a joint and coordinated manner [2].

2.3 Scenario-based verification and validation

Significant efforts have been dedicated to developing validation and assessment methodologies for ADS. However, a harmonized method that comprehensively addresses verification, validation, and certification requirements across various automation levels and feature classes remains elusive. A feature class denotes groups or families of automated driving functions that differ in scope, complexity, and intended use, used to distinguish between types of automation features that carry different verification, validation, and certification demands.

A critical aspect of ADS evaluation is the definition of transparent and explicit evaluation targets, supported by evidence addressing both safety and security. One proposed approach is the use of a quantitative risk norm (QRN) [51], as introduced in ISO TS 5083. The QRN defines consequence classes, each associated with an upper frequency bound for the occurrence of the respective consequences. Compared to the coarser risk calibration employed in ISO 26262, the QRN provides a more fine-grained and quantitatively oriented means of demonstrating goal fulfillment.

Currently, scenario-based testing for automated driving is gaining prominence, building on Road vehicles - Test scenarios for automated driving systems ISO 3450X [11, 12, 13], which in turn draws on scenario-based principles originally introduced by Cem Kaner [52] and later adapted to the automotive domain by Ulbrich et al. [53]. Nevertheless, determining whether a test suite sufficiently covers the test space remains a challenging task. [54, 55] Developing systematic methods to align scenario requirements with distinct test environment capabilities is part of the solution. Integrating ODD parameters with test attributes can address this gap by enabling the automated allocation of test cases to the appropriate test environments. A subcategory of this topic is the external assessment of the appropriateness of such allocations, as required by functional safety standards. This assessment is similarly complex due to challenges in defining scope and linking the intended context to the test environment appropriateness.

2.4 Multi-pillar approach

The NATM validation approach, suggested by VMAD, adopts a multi-pillar foundation for the validation of ADS, composed of a scenarios catalog and five validation methodologies, referred to as pillars. The pillars have different meanings and significance in the suggested certification process. Figure 2.1 shows a graphical interpretation of the proposed New Assessment / Test Method for Automated Driving (NATM) [18].

The Assessment of Cybersecurity-informed Safety (AoCiS) proposed in [5] begins by defining joint goals and validating achievement by providing evidence that the system is safe even when exposed to security threats. The observable boundary of the testing can be expanded to include human-machine interaction validation procedures [50]. The assessment results should be helpful as a complement to the proposed multi-pillar approach, forming a comprehensive assessment and contributing to an evaluation of cybersecurity-informed safety concerns that have been systematically unaddressed hitherto. The AoCiS is designed to be a transparent validation procedure,

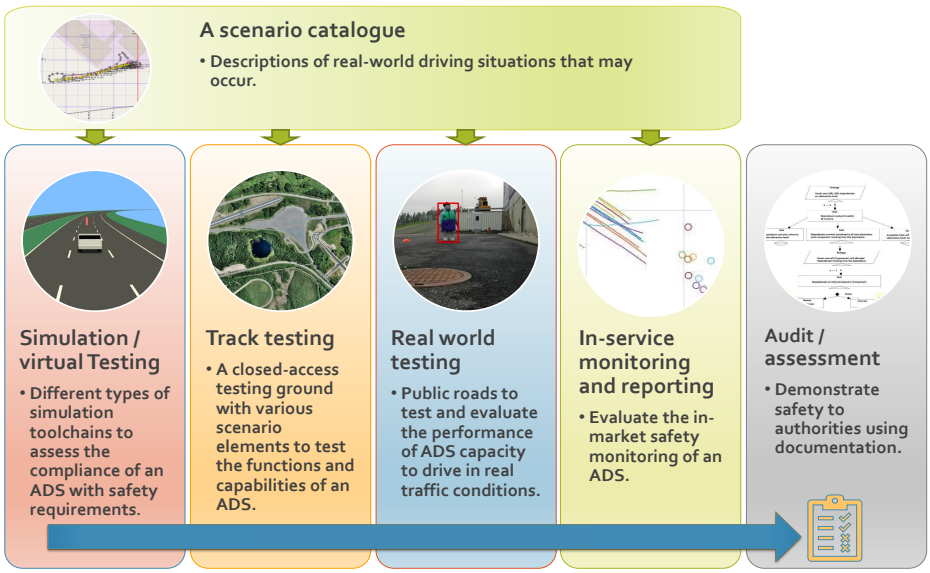


Figure 2.1: Image of proposed NATM process[a] with a multi-pillar approach for the validation of ADS, composed of a scenarios catalog and five validation methodologies.
[a]Inspired by content in [18]

harmonized with the proposed NATM multi-pillar approach; the result is expected to be seamlessly integrated into a comprehensive assessment. The AoCiS audit and assessment scrutinize firstly if applied processes and methods in system development relevant for cybersecurity-informed safety are adequate, complete, and consistent, and secondly, assesses the results of accelerated testing, such as virtual verification [56], which produces evidence that supports the assertion that the ADS behavior is safe and secure.

2.5 Co-assessment

The aggregated evidence from all previous safety and security-related activities, which proves that the system is sufficiently safe and secure, will need to be independently assessed. ISO 26262 emphasizes the need to consider interdependency with cybersecurity; however, there is no support for transforming these requirements into a hands-on assessment process for conformance to ISO 21434. This is also true for ISO 21448, which states that the results of SOTIF activities can be taken into account in ISO 26262's functional safety assessment without further elaboration. ISO 21434 addresses cybersecurity assessment separately; however, the exact connection to ISO 26262 assessments is a non-normative annex. For example, one connection is that safety risks are best defined and collected within the context of ISO 26262. All evaluations must use the scheme of independence as outlined in ISO 26262. However, independently

conducting safety and cybersecurity assessments renders the evaluation of interactions between these concerns challenging, if not impossible. The synchronized development process proposed aims to facilitate a joint, incremental assessment, potentially enhanced through the implementation of assurance case templates [57]. Despite this, current standards and regulations offer insufficient guidance on performing a comprehensive co-assessment and detailing its implications. Yet, essential prerequisites-such as a joint definition of safety and security, and the establishment of transparent, combined development goals-can be addressed through a well-synchronized process.

Certification practices also face substantial challenges in assessing ADS, where physical testing methods used traditionally for safety certification become infeasible due to the complexity and high level of autonomy, especially at SAE levels L4 and L5. Cybersecurity threats directly impact safety, highlighting the urgency for integrated assessment procedures that account for the dynamic and interconnected nature of ADS. Current certification frameworks and standards, including ISO 21434 [44] and UNECE regulations R155 [45] and R156 [58], remain generic and inadequately tailored to the nuanced distinctions and heightened risks associated with high levels of automation. Furthermore, the evolving nature of cyber threats introduces significant complexity and uncertainty into certification processes [59].

Here, the focus lies on how an independent third party can supply the technical assessment service, continuously conducted during product development, to support conformance with all applicable standards. The product and process assessment methodology should be reinforced by targeted physical and virtual tests (see Figure 2.1), which are scalable to cover the comprehensive variety of feature classes with minimal insight into the implementation. The somewhat neglected area of safeguarding against cyber-attacks to ensure the system's safety and integrity is particularly interesting for the targeted additional tests.

Before releasing an ADS to the market, it must be shown that cybersecurity threats do not jeopardize safety. As it is virtually impossible to validate an automated vehicle against all possible scenarios it will face in the real world, least of all in combination with security threats, there is a need to strike a balance between the representativeness of the tests and the reliability of the performance indicators [60]. One way is to define testing and validation procedures for ADS features that can leverage tests in both simulation and real environments, such as test tracks, to conduct a comprehensive assessment. An approach with a similar goal generates meaningful attack vectors on a system. It executes cybersecurity test cases on a black box using pattern-matching-based binary analysis and translation mechanisms to formal attack descriptions and model-checking techniques [61]. The scope of our investigation is thus limited to assessment testing, where a small test batch must be shown to represent a whole class of ADS features. Firstly, the focus is on setting up an approach to generate valid test scenarios for cybersecurity applicable to classes of ADS. The intention of this work is not primarily to support the implementation of a comprehensive ADS, which is a much more complex undertaking, but rather to facilitate the investigation of the interplay between safety and security by providing guidance and inspiration for independent assessment tests.

Chapter 3

Research Summary

This chapter presents the research methodology and process employed, which follows the Design Science Research (DSR) paradigm. The focus is on the iterative development and refinement of a single artifact-the Safety Assurance Framework (SAF)-aimed at enhancing the assessment of ADS. Section 3.1 outlines the methodological foundation and the overall research design up to the identification of the general research goal. The specific research questions are addressed separately in Section 3.2, followed by an introduction to the research process in Section 3.3.

3.1 Research Methodology

Research methodology refers to the systematic approach employed to investigate a research problem scientifically. It encompasses the principles and procedures by which researchers conceptualize, design, and conduct studies to generate valid and reliable knowledge.

The research presented in this thesis falls within the discipline of information systems, which often involves creating artifacts to address practical problems. In this context, the value of research is determined by its relevance and the contribution it makes to the field.

DSR is a research paradigm that emphasizes the creation and evaluation of artifacts designed to solve identified problems. It seeks to extend human and organizational capabilities by developing innovative artifacts [62]. DSR is applicable across various disciplines, including computer science and information systems [63, 64].

Artifacts in DSR can take various forms, including algorithms, models, methods, and frameworks. They are considered carriers of knowledge, encapsulating design principles and insights that can be applied in practice. The effectiveness of DSR is not solely determined by the artifact itself but also by the rigor of its development and evaluation processes [65].

Hevner et al. [62, 65] propose a three-cycle view of DSR, comprising the Relevance Cycle, the Design Cycle, and the Rigor Cycle, see Figure 3.1. The Relevance Cycle connects the research to the application environment, ensuring that the artifact addresses

real-world needs and requirements. The Design Cycle focuses on the iterative process of building and evaluating the artifact. The Rigor Cycle links the research to the existing knowledge base, grounding the artifact in established theories and methods.

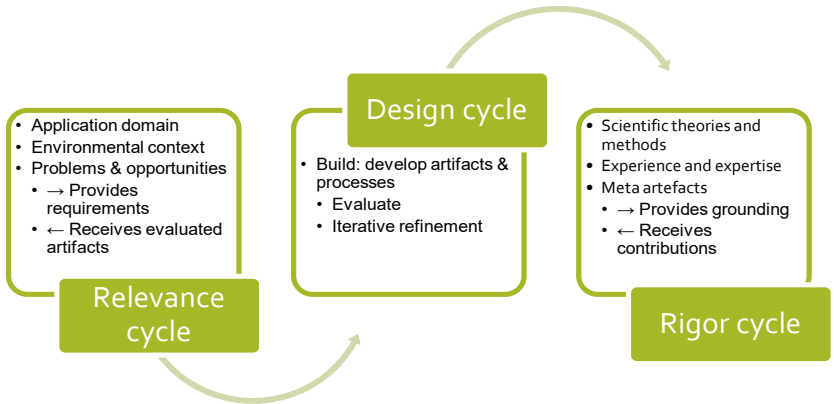


Figure 3.1: Design Science Research Cycles.

The iterative nature of DSR enables continuous refinement of the artifact, thereby enhancing its relevance and effectiveness. By integrating the three cycles, researchers can ensure that the artifact is both practically useful and theoretically grounded.

In this thesis, DSR is employed as the overarching methodology for developing the SAF. The research process involves iterative cycles of design and evaluation, informed by both practical considerations and theoretical insights. This approach facilitates the development of a robust framework for assessing the safety of ADS.

3.2 Research Process

The research process adheres to the DSR methodology outlined in Section 3.1, centering on the iterative development and refinement of a single artifact-the SAF. This process is visually represented in Figure 3.2, which provides an overview of the SAF and its assessment components.

The initial phase involves conceptualizing the SAF, drawing upon existing standards such as ISO 26262 [40], and identifying gaps in current assessment practices. This phase corresponds to the Relevance Cycle, where the problem context and requirements are established.

Subsequent iterations involve designing and refining the SAF, incorporating feedback from stakeholders and aligning with theoretical frameworks. These activities align with the Design Cycle, emphasizing the construction and evaluation of the artifact.

Throughout the process, the research is informed by existing literature and best practices, ensuring that the SAF is grounded in established knowledge and expertise.

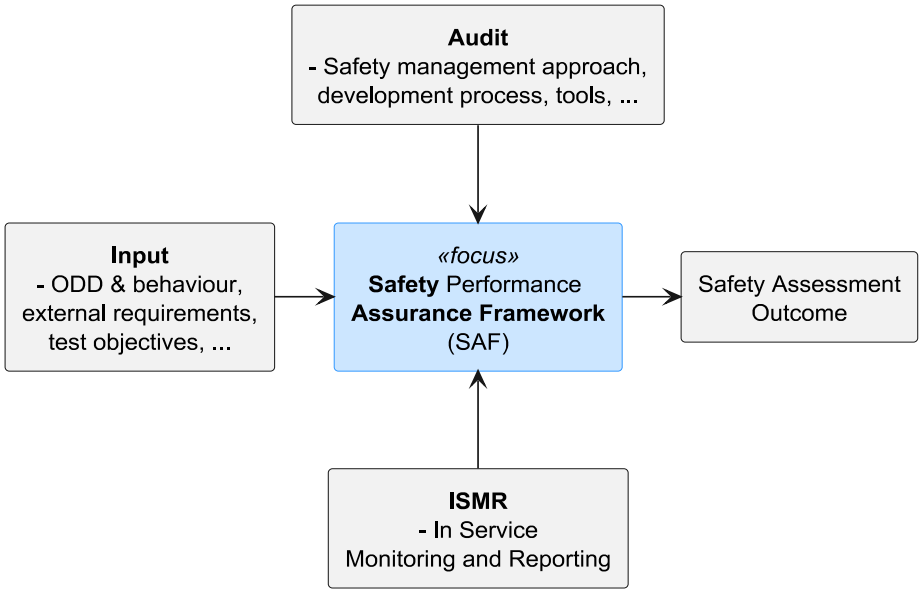


Figure 3.2: Context of the Safety Assurance Framework to be incrementally improved.

This aspect corresponds to the Rigor Cycle, linking the research to the broader academic discourse.

The overarching goal of the research is to develop a comprehensive and adaptable framework for assessing the safety of ADS. By employing the DSR methodology, the research aims to produce an artifact that is both practically relevant and theoretically sound, contributing to the advancement of safety assessment practices in the field of ADS.

3.3 Research Questions

The scope demarcations and research motivation are presented here. The research is based on several assumptions (see Section 2). The primary objective of this research is to clarify the role of independent assessment in addressing the challenges of establishing confidence and user awareness in CCAM systems.

The focus is on safety, security, and the interplay of these quality attributes in ADS. The main challenge for assessment is the complexity of the systems and the multitude of driving situations and scenarios that must be validated, depending on the operational domain. Testing an ADS-equipped vehicle alongside other actors is both challenging and hazardous. Currently, available processes do not fully address the need to simultaneously define and ensure the fulfillment of multiple quality attributes under these conditions. A well-thought-out definition of safety, security, and their interplay is expected to support making and assessing a joint argumentation that the evidence for

achieved safety of the item to a transparent assurance target is sufficient.

The research questions have been formulated to enable the objective assessment of automated systems and the determination of levels of compliance, while conforming to applicable standards and guidelines for an ADS in the automotive domain. The assessment results should be helpful as a complement to the proposed multi-pillar approach, forming a comprehensive assessment and contributing to an evaluation of cybersecurity-informed safety concerns that have been systematically unaddressed hitherto.

RQ1 What are the specific safety and security-informed safety requirements applicable to key enabling technologies essential for ADS validation, with particular emphasis on perception, communication, and positioning?

RQ2 How can the relationship between safety and security-informed safety be systematically characterized and jointly managed throughout ADS development?

RQ3 How can an independent agency systematically assess ADS's safety and security-informed safety through structured scenario-based testing?

These questions guide the iterative development of a central research artifact, the SAF, which integrates multiple sources of information and evaluation mechanisms, as shown in Figure 3.2. The SAF is refined through a design science approach, as illustrated in Figure 3.3, where each cycle contributes to aligning the framework with the research questions and the operational needs of independent assessment.

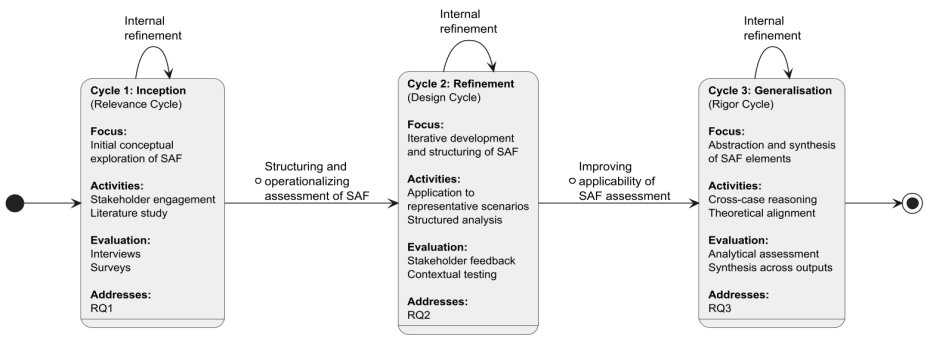


Figure 3.3: Iterative refinement plan for the SAF artifact through the DSR

The DSR process is structured into three interlinked cycles that correspond directly to the research questions. In the first cycle, which represents the inception phase and corresponds to the Relevance Cycle, the conceptual basis of the SAF is explored through engagement with stakeholders and a systematic literature review. The evaluation methods include interviews and surveys. This cycle addresses the first research question by identifying the relevant safety and security-informed safety requirements associated with enabling technologies for ADS validation.

The second cycle focuses on refinement and corresponds to the Design Cycle. During this phase, the SAF is iteratively developed and structured, tested against representative scenarios, and evaluated through stakeholder feedback and contextual analysis. This cycle addresses the second research question by providing a structured approach to jointly manage safety and security-informed safety concerns throughout the ADS development process.

The third cycle involves generalization and corresponds to the Rigor Cycle. Here, the framework components are abstracted and synthesized through cross-case reasoning and theoretical alignment. Evaluation in this phase includes analytical assessment and integration of results across outputs. This cycle addresses the third research question by examining how independent agencies can conduct systematic assessments of ADS using structured scenario-based methods.

Each DSR cycle thereby contributes to the iterative maturation of the SAF, aligning its structure and logic with the underlying research questions. The framework remains semi-mature, with the NATM [18] offering conceptual guidance that requires further development before it can become a fully operational methodology. While the individual components are relatively well established, their integration and application form the central subject of this research.

Chapter 4

Contributions

The research presented in this thesis is structured as a compilation of interconnected contributions, each providing distinct insights into the overall study objective. Section 3 previously described the research process, elucidating the interplay between research questions and the iterative design science research cycles. Section 4.1 systematically presents these contributions, their motivational foundations, and their mutual dependencies. Brief summaries of each associated publication are included in Section 4.2. Finally, Section 4.3 elaborates on the evaluation methods applied to ensure the utility and methodological rigor of the developed artifacts.

4.1 Scientific Contributions

This section presents and discusses the scientific contributions derived from the published works included in this thesis. The research was guided by overarching research questions, and each contribution provides partial yet significant insights toward addressing these inquiries. Collectively, these contributions represent structured approaches to advance safety and security assessment practices for ADS. However, they do not constitute exhaustive solutions, nor do they claim exclusivity in addressing the research challenges posed.

The research questions are addressed through eight contributions. These are grouped into three clusters: identifying context-dependent requirements (C1, C7), managing multi-concern development (C2–C5), and supporting independent assessment through scenario-based testing (C6, C8).

Each of these contributions addresses specific aspects of the refined research questions, providing coherent methodological and conceptual advancements for assessing the integrated aspects of safety and cybersecurity within ADS development.

The first concerns the requirements for safety and security-informed safety, with a focus on key enabling technologies in ADS, including perception, communication, and positioning. This question led to the development of Key Enabling Technologies (KETs) assessment templates (C1) and the formalization of the operational design domain using a specialized language (C7).

Table 4.1: Mapping between papers and research contributions.

Paper	ID	Contribution Description
A	C1	Development of assessment templates for Key Enabling Technologies (KETs), such as communication, positioning, and cybersecurity, linked to the ODD, demonstrated through a remotely assisted truck use case [1].
B	C2	Comparative analysis of separate and co-engineered multi-concern development approaches, focusing on safety and cybersecurity, highlighting synergies in design and verification phases using a positioning component use case [2].
C	C3	Proposal of structured argument patterns to support simultaneous assurance of safety, cybersecurity, and availability in Connected Automated Driving (CAD) systems [3].
D	C4	Introduction of a unified process model delineating relationships and scheduling between work products for safety and security implementation within the current standardization landscape [4].
	C5	Development of a synchronization mechanism within the V-model framework to systematically align safety and security considerations throughout the development lifecycle [4].
E	C6	Method for assessment of cybersecurity-informed safety (AoCiS) leveraging simulation-based and real-world tests to evaluate the interplay between security and safety, thus mitigating associated risks [5].
F	C7	Formalization of the ODD using the Pkl language to address specification challenges and enhance usability through specialized configuration language features [6].
G	C8	Methodology for evaluating the suitability of test case allocation to various test environments [7], based on an extended formalized ODD incorporating key testing attributes, demonstrated through an automated reversing truck function case study.

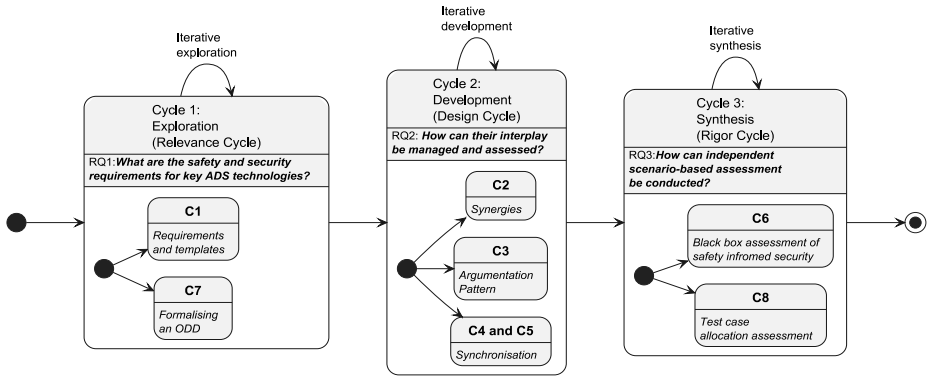


Figure 4.1: A graphical representation of research cycles and contributions.

The second research question addresses how safety and security-informed safety can be jointly managed during the system development process. This resulted in the analysis of multi-concern development approaches (C2), the development of argument patterns for multi-concern assurance (C3), and a unified process model for safety and security integration (C4). A synchronization mechanism aligned with the V-model was also proposed to support consistent coordination of these concerns (C5).

The third research question examines how an independent agency can evaluate systems with intertwined safety and security requirements through scenario-based testing. This is addressed through a method for assessing cybersecurity-informed safety (C6) and a methodology for evaluating test case allocation based on a formalized operational design domain (C8).

By addressing context derivation, process integration, and test allocation, the contributions form a coherent framework for the structured evaluation of safety- and security-related claims in ADS. They support systematic and practical assessment approaches that account for technical aspects of concerns. Figure 4.1 shows the conceptual structure and relational mapping of the contributions, respectively. These figures support the traceability from research questions to artifacts and their application.

The scope of these contributions is limited to methodological development and structured technical assessment support. Integration with regulatory approval processes and complete certification toolchains is outside the scope of this work. Each contribution is linked to the corresponding publication, as described in Section 4.2.

4.2 Included publications

This section lists and presents the research publications constituting the thesis arrangement as a collection of published works.

4.2.1 Paper A

Title: Enhancing Safety Assessment of Automated Driving Systems with Key Enabling Technology Assessment Templates.
Authors: Martin Skoglund, Fredrik Warg, Anders Thorsén and Mats Bergman.
Proceedings: In *Vehicles*. 2023. MDPI.
Contribution: The main contribution of this paper is the proposal and demonstration of a method to support the safety assessment of Automated Driving Systems (ADS) by introducing Key Enabling Technology (KET)-specific assessment templates. These templates complement existing scenario-based evaluation frameworks by incorporating technology-aware guidance. The proposed method structures the derivation of assessment attributes and criteria by analyzing system requirements and mapping them to ODD conditions, with a specific focus on communication, positioning, and cybersecurity.

The assessment templates are developed through a systematic process that involves collecting requirements, analyzing technology reliance, and refining them into performance indicators and functional scenarios. By doing so, the method enhances the traceability, relevance, and completeness of safety assessments, aligning them with future type-approval processes and standards. The use of these templates is illustrated through a case study involving a remotely assisted truck in a constrained urban ODD, where evaluation scenarios were derived for both positioning and 5G-based communication features. The method demonstrates how KET-awareness can be operationalized into structured, repeatable assessments (C1). The scope is limited to supporting the evaluation of ADS safety claims by providing guidance on assessable attributes, rather than replacing system-level assessment procedures.

My contribution: I was the main driver of the paper, aggregating and evaluating the collected requirements and writing the bulk of the text. I played a central role in the project effort to collect and aggregate information through surveys and interviews, which was then used to compose a specification for testing, focusing on the three key enabling technologies defined. Conceptualization and methodology, M.S.; validation measurements, M.B.; investigation, M.S., A.T., and F.W.; writing—original draft preparation, M.S., A.T., and F.W.; writing—review and editing, M.S., A.T., and F.W.; supervision, F.W. All authors have read and agreed to the published version of the manuscript.

4.2.2 Paper B

Title: In search of synergies in a multi-concern development lifecycle: Safety and cybersecurity.
Authors: Martin Skoglund, Fredrik Warg, Behrooz Sangchoolie.

Proceedings: International Conference on Computer Safety, Reliability, and Security, SAFECOMP Workshops 2018.
Contribution: We investigated and compared two approaches to multi-concern development: separate and co-engineered. Quality attributes (concerns) were analyzed sequentially, whereas the latter were analyzed in parallel. The quality attributes analyzed were functional safety and cybersecurity, and the analysis was performed in accordance with the ISO 26262 standard (addressing functional safety) and the first working draft of the ISO/SAE 21434 standard (addressing cybersecurity for road vehicles). In this context, a situated implementation of a use case, involving the development of a positioning component, was used to explore potential synergies.

The main contribution of our analysis shows that on the design side of the development lifecycle, the significant gain is the completeness of the analysis when considering both attributes together. Still, the shared activities overlap is small (C2).

My contribution: I was the lead author of the paper, responsible for formulating the research problem, defining the two approaches, developing and conducting the comparative evaluation, and drafting the majority of the manuscript. I also presented the work. The co-authors contributed primarily through iterative review, constructive feedback, and refinement of the quality of the results.

4.2.3 Paper C

Title: Argument Patterns for Multi-Concern Assurance of Connected Automated Driving Systems.
Authors: Fredrik Warg, Martin Skoglund.
Proceedings: 4th International Workshop on Security and Dependability of Critical Embedded Real-Time Systems, CERTS 2019.
Contribution: In this paper, we propose a pattern to aid in the structured argument for the simultaneous fulfillment of attributes such as safety, cybersecurity, and availability, which is helpful as a core part of an assurance case related to an ADS.

The proposed argument pattern (C3) is a model that contains statements expressing relationships for simultaneously covering the dimensions of *(1) multiple concerns*, *(2) standards conformance*, *(3) element-out-of-context*, and *(4) system lifecycle* in an argument for a multi-concern assurance case. The argument pattern is modeled in goal structuring notation (GSN). We illustrate the use of this argument pattern in a case study relevant to ADS.

A danger with such multi-dimensional arguments is that they become unwieldy and incomprehensible, thus failing to fulfill their primary purpose, which we attempt to overcome with a clear and consistent structure.

My contribution: The concepts presented in the paper are a product of my and Fredrik Warg's iterative collaboration on modeling argumentation. The primary author, Fredrik Warg, wrote most of the text and presented the work.

4.2.4 Paper D

Title: Synchronisation of an Automotive Multi-Concern Development Process.

Authors: Martin Skoglund, Fredrik Warg, Hans Hansson, Sasikumar Punnekkat.
Proceedings: 40th International Conference on Computer Safety, Reliability, and Security, SAFECOMP Workshops 2021.
Contribution: In this paper, we propose a unified model of processes that express relationships and scheduling between work products, supporting the implementation of safety and security, and their interaction within a development procedure that operates within the current standardization landscape (C4).

An additional contribution to the unified process map, organized around a generic V-model, apart from containing applicable processes, artifacts, and activities, also describes a mechanism for synchronizing content and scheduling between concerns at each step (C5). The synchronization mechanism is a framework artifact that advances conceptual guides to support alignment between the concerns in a multi-concern process.

My contribution: I was the main driver of the paper, including formulating the problem and defining, developing, and evaluating the approach. The co-authors contributed with improvements and overall guidance.

4.2.5 Paper E

Title: Black-Box Testing for Security-Informed Safety of Automated Driving Systems.
Authors: Martin Skoglund, Fredrik Warg, Hans Hansson, Sasikumar Punnekkat.
Proceedings: Proceedings of 93rd Vehicular Technology Conference, VTC spring 2021.
Contribution: The primary contribution of this paper is the proposed method for assessing Cybersecurity-informed Safety (AoCiS). The work is based on input from previous findings. It describes the steps that can be used to assess security-informed safety by leveraging tests in both simulation and real environments, while reducing the associated risks.

The method outlines an approach to generate valid and relevant test scenarios for cybersecurity that apply to classes of automated driving systems, supporting the investigation of the interplay between security and safety by providing guidance and inspiration for independent assessment tests (C6). The scope of our investigation is thus limited to assessment testing, where a small test batch can be shown to represent a whole class of ADS features.

My contribution: I was the main driver of the paper, including formulating the problem and defining, developing, and evaluating the method. The co-authors contributed with improvements and overall guidance.

4.2.6 Paper F

Title: Formalizing Operational Design Domains with the Pkl Language.
Authors: Martin Skoglund, Fredrik Warg, Hans Hansson, Sasikumar Punnekkat.
Proceedings: *Accepted to the 36th Intelligent Vehicle Symposium*. IV 2025.
Contribution: The main contribution of this paper is the formalization of ODDs using the Pkl configuration language. This work addresses the limitations of existing

ODD representations by proposing a method that supports machine-readable, modular, and extensible ODD definitions. The approach is grounded in current standards and practices, notably ISO 34503, and aims to improve consistency, traceability, and validation of ODDs in the context of automated driving systems.

The method provides a structured yet flexible way to specify and instantiate ODDs while enabling integration with testing and safety assurance processes. The Pkl-based formalization aligns with identified assessment criteria and facilitates reuse, scenario coverage analysis, and dynamic test process integration. While the work does not aim to define a new taxonomy, it advances the usability and management of ODD specifications by enhancing configuration integrity and enabling seamless toolchain compatibility (C7). The scope of this work is limited to the formal representation and management of ODDs as safety artifacts.

My contribution: I was the primary driver of the paper, including formulating the problem, defining, developing, and evaluating the method. The co-authors contributed with improvements and overall guidance.

4.2.7 Paper G

Title: Methodology for Test Case Allocation based on a Formalized ODD.
Authors: Martin Skoglund, Fredrik Warg, Hans Hansson, Sasikumar Punnekkat.
Proceedings: *Submitted 20th International Workshop on Dependable Smart Embedded Cyber-Physical Systems and Systems-of-Systems*. DECSoS 2025.

The main contribution of this paper is the proposal and demonstration of the Methodology for Test Case Allocation based on a Formalized ODD (METAFODD), which enables the systematic allocation of test cases to suitable test environments by extending a formalized ODD with test environment attributes. This work extends a previously established machine-readable ODD representation with additional test environment attributes relevant to safety assessment. By incorporating elements such as safety hazard mitigation, test complexity, and fidelity of both the test environment and the system under test, the method enables systematic evaluation of whether a given test environment is suitable for executing a particular scenario.

The proposed allocation approach supports automation by enabling structured comparison between scenario requirements and test environment capabilities. It is implemented using the Pkl configuration language, which allows reuse and extension of ODD specifications and supports integration with validation toolchains. The applicability of the method is illustrated through a reversing-truck case study that demonstrates the need for environment-specific fidelity when evaluating sensor performance under challenging conditions, such as sun glare. This method provides a foundation for scalable safety testing in automated systems and contributes to the integration of scenario-based assessment with formal ODD definitions (C8). The scope of the contribution is limited to test case allocation based on ODD conformance, without addressing broader aspects such as scheduling or cost optimization.

My contribution: I was the primary driver of the paper, including formulating the problem, defining, developing, and evaluating the method. The co-authors contributed with improvements and overall guidance.

4.3 Contribution Evaluation

The Safety Assurance Framework (SAF) artifact has been refined iteratively, with evaluation grounded in its intended use for structured technical assessment. It is through this application-particularly in the context of safety and security evaluation for ADS-that the artifact's utility and contribution can be meaningfully assessed. Evaluation serves not only to demonstrate utility but also to uncover situational constraints, thereby informing further refinement and contributing constraint knowledge back to the evolving SAF artifact.

Evaluations of the contributions were conducted through case studies, experimental demonstrations, and structured analyses, depending on the nature of each contribution. C1 was assessed in a real-world use case with on-site measurements within a constrained operational domain. The evaluation indicated improved efficiency in assessment activities, while also highlighting limitations related to scenario completeness, the need for regular updates, and the benefit of integrating multi-concern perspectives from the outset. Further evaluation, as reported in [66], complements this by applying the method across simulation, scaled physical models, and full-size vehicles in a reverse parking scenario. The results demonstrated consistent performance across environments, strengthening the evidence of feasibility, but also revealed limitations related to sensor uncertainty, positioning accuracy, and reduced reliability in edge or non-nominal conditions.

C2 was examined through a case study of joint development. The results showed increased completeness in co-design and reuse potential in testing activities. C3 was applied to a use case involving multiple interdependent concerns. The structured approach improved traceability and facilitated the management of complexity, although tool support was identified as necessary for scalability to larger systems. C4 and C5 were evaluated through analysis of standard outputs, which revealed relationships between processes. The results suggest that aligning outputs can enhance coordination and consistency; however, they remain theoretical and have yet to be validated in practice.

C6 was evaluated using a representative example. The formalization met several criteria related to structure and toolchain integration; however, the evaluation also identified constraints, such as the need for translation between representations and the addition of logic to enable automation. Additional evaluation in [66] demonstrated how internal system conditions, environmental properties, and test environment capabilities can be systematically connected across simulation, scaled physical setups, and full-scale tests. This reinforced the practical value of the approach but also underlined limitations, including measurement uncertainties, reduced repeatability in physical environments, and challenges in aligning model fidelity across different testbeds.

C7 was tested in an experimental setup involving a targeted disruption. The method successfully generated relevant scenarios, although the scope was limited to a single threat category, and further investigation is required to establish the reliability of simulation results. C8 was assessed in a use case involving automated maneuvering. The initial evaluation demonstrated that scenario allocation based on environmental properties is feasible and scalable; however, its current applicability is constrained by

its reliance on a preliminary classification scheme and is dependent on the specific allocation purpose. The evaluation presented in [66] extends these findings by showing that scenario allocation grounded in a formalized ODD can be operationalized across heterogeneous test environments in the reverse parking use case. The results indicated that the approach is feasible and supports coverage analysis, but also highlighted limitations stemming from restricted ODD scope, dependencies on well-defined KPIs and acceptance thresholds, and environment-specific constraints that limit the execution of certain scenarios.

The evaluation is informed by the knowledge contribution framework proposed by Gregor et al. [67], which provides a basis for classifying and interpreting the nature and maturity of design contributions. Furthermore, the assessment is structured around the seven design science research guidelines formulated by Hevner et al. [62], as summarized in Table 4.2. Although originally developed for the information systems domain, these guidelines are broadly applicable to research concerned with the development of innovative and purposeful artifacts in complex problem settings.

Table 4.2: Design-Science Research Guidelines

Guideline	**Application in Thesis**
Design as an Artifact	Contributions are artifacts in the form of constructs (C1, C2, C4, C5), a model (C3), and methods (C6, C7, C8). Artifacts include the formalization of ODD and methodologies for structured scenario generation and test allocation (C7, C8).
Problem Relevance	The relevance and importance of the contributions have been clarified by explicitly linking requirements to defined ODD and systematically addressing the interplay between safety and security within ADS, as presented in Chapter 2.
Design Evaluation	The utility, quality, and efficacy of the contributions have been evaluated explicitly in the context of formalized ODDs and structured scenario-based methods, described in Part II and Section 4.3.
Research Contributions	Clear and refined contributions aligned with the adjusted research questions (RQ1, RQ2, RQ3) are defined explicitly in Chapter 4, with improved integration of formalized ODD descriptions and structured methods for ADS assessments.
Research Rigor	Contributions have been systematically constructed and linked to a clearly defined ODD context. Further validation is recommended for the structured test scenario methods (C7, C8) to enhance rigor.
Design as a Search Process	The iterative research approach, including multiple reframing steps to address the complexity of jointly assessing ADS safety, security, and ODD interactions, has been maintained and enriched, as outlined in Chapter 3.
Communication of Research	The contributions have been disseminated, peer-reviewed, and presented to relevant practitioners, now further enriched by explicitly incorporating formalized ODD and structured scenario-based testing, as listed in Part II.

The contributions are evaluated and categorized according to their applicability or the knowledge they generate (see Figure 4.2). After evaluation, each contribution is assigned a classification to reflect the nature of the knowledge provided. This categorization facilitates comparison and helps the reader understand the type of knowledge contribution made by each effort.

As per the Gregor and Hevner framework, contributions such as C1, C2, and C5 align with the exaptation category. These contributions represent innovative repurpos-

ing of existing knowledge structures to address emerging challenges in ADS assurance, where the maturity of the application is still low and the domain-specific adaptation is novel. The evaluation of safety and security requirements reflects the instantiation of practical necessities in a specific operational setting, drawing on knowledge collected from active stakeholders involved in ADS development and assessment. The aggregation and contextualization of this knowledge were crucial in supporting harmonized European testing approaches for automated transportation.

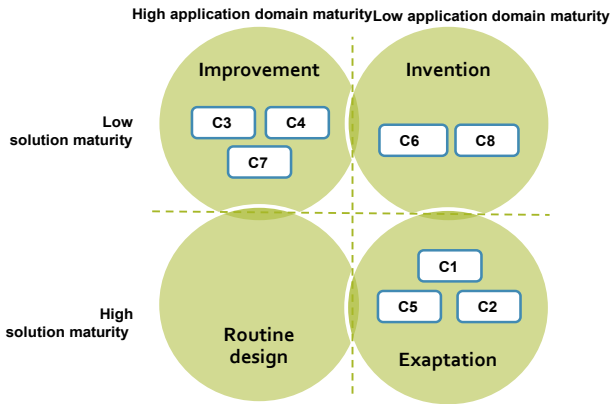


Figure 4.2: Design Science Research Knowledge Contribution Framework.

Structured argumentation is extended and refined through the development of a novel pattern addressing safety, security, and their overlap jointly (C3). A detailed understanding of the applicable standardization landscape enabled the design of artifacts that integrate multiple processes for conformance (C4). The justification for jointly addressing safety and security throughout the development lifecycle is based on completeness-ensuring all concerns and their interdependencies are consistently treated across phases-which is argued to improve over traditional separated approaches.

The formalization of ODD using the Pkl configuration language (C7) is categorized as an improvement. This contribution enhances existing practices by increasing precision, modularity, and integration capability, thereby improving the management and reuse of ODD specifications. Its added value lies in advancing the usability and effectiveness of a foundational artifact for ADS safety assurance.

The method for independent assessment of security-informed safety using tests in simulation and real environments (C6) falls within the invention quadrant. Similarly, the methodology for test case allocation based on a formalized ODD (C8) is categorized as an invention. Both contributions introduce novel methods to the domain, with partial validation in illustrative settings. While they remain in the early stages of maturity, their potential impact justifies their classification as original design contributions. As noted by Gregor and Hevner [67], the creation of new knowledge must begin with innovation, even when completeness and generalizability are still evolving.

Chapter 5

Summary and Conclusion

This chapter summarises how the work presented in Section 4.2 contributes to the development of an integrated Safety Assurance Framework (SAF) for the assessment of ADS. The high-level schematic process blocks of such an SAF are illustrated in Figure 5.1.

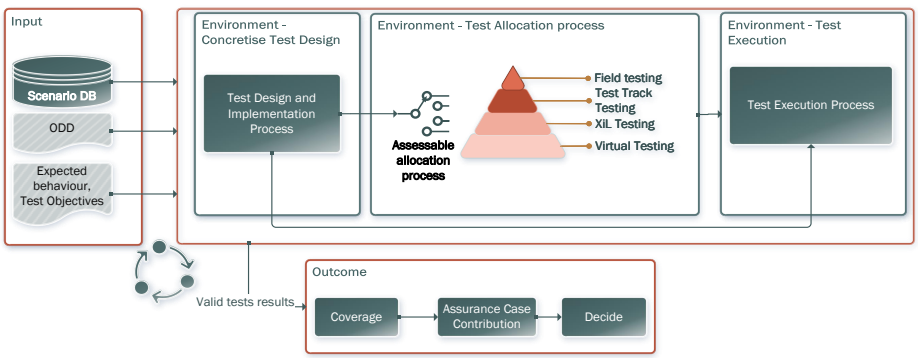


Figure 5.1: The high-level process blocks of a SAF.

The specific mapping of contributions to the schematic SAF process is presented in Table 5.1.

Table 5.1: Influence of research contributions on high-level SAF process blocks.

High level SAF process blocks	C1	C2	C3	C4	C5	C6	C7	C8
Input	x	x	x	x	x		x	
Test Environment: Test design, allocation and execution		x	x	x	x	x	x	x
Outcome			x			x		x

The thesis addresses the construction and evaluation of key artifacts and concepts

necessary to support the independent assessment of safety-informed security. The objective has been to enable structured and context-aware evaluation by leveraging operational design domains, requirements for key enabling technologies, and multi-concern analysis. Particular emphasis has been placed on combining simulation and real-world testing in a way that mitigates the associated risks.

Proving top-level safety claims through scenario-based testing is independent of the underlying implementation technology and therefore remains agnostic to AI. The articulation of assessment aspects on modern developments, such as foundation models and end-to-end approaches, is instead located in other parts of a broader assessment framework that extends beyond the direct performance evaluation of ADS. Implementation-dependent aspects, such as the data lifecycle that drives AI development, are addressed in the Audit part of the SAF (Figure 3.2), while claims based on design assumptions requiring continuous monitoring during operation are considered in the in-service monitoring part (Figure 3.2). Together, these components provide the broader assessment context in which scenario-based testing, as depicted in Figure 5.1, is situated.

The mapping in Table 5.1 is structured according to the three principal process blocks of the SAF: Input, Environment, and Outcome.
The **Input block** is influenced by contributions related to the identification and formalization of requirements and foundational arguments. Specifically, the development of assessment templates for key technologies (C1), the analysis of multi-concern synergies (C2), the establishment of argumentation patterns (C3), the alignment of these patterns with existing development processes (C4), the definition of a synchronization mechanism (C5), and the formalization of the operational design domain (C7) collectively inform the structured and transparent specification of assessment criteria.

The **Environment block** is shaped by contributions that enable the operationalization of requirements and argumentation into testing processes. The synergy analysis (C2), argumentation approach (C3), process alignment (C4), and synchronization mechanism (C5) all support the coordinated design and implementation of test activities. The introduction of a method for cybersecurity-informed safety assessment (C6), the formalization of the ODD (C7), and the development of a methodology for test scenario allocation (C8) further contribute to the practical execution and allocation of scenario-based testing in both simulated and real environments.

The **Outcome block** reflects the structured evidence generated through the integrated assessment process. Contributions relating to argument patterns (C3), the cybersecurity-informed safety assessment method (C6), and the scenario allocation methodology (C8) collectively support the production of traceable, context-aware results, ensuring that validation activities address both safety and security requirements.

5.1 Summary

This thesis posits cybersecurity-informed safety as a validation strategy that integrates structured requirements, joint argumentation, and scenario-based evidence generation. The proposed SAF artifact now includes several incremental improvements in response to the research questions, serving as a foundational element that connects early design

considerations with later-stage assessment activities, as illustrated in Figure 5.1 and Table 5.1. In particular, the work shows how operational design domains, key enabling technology assessments, and multi-concern argumentation can be systematically combined to form a coherent and transparent safety assurance process. Rather than being stand-alone solutions, the proposed sub-processes are designed as building blocks within an extensible architecture that can adapt to emerging technologies and evolving regulatory needs.

Taken together, the contributions establish a solid foundation for independent, transparent, and context-aware evaluation of ADS. They advance current practice in safety and security assurance by providing structured methods for capturing requirements, developing argumentation, and conducting scenario-based validation. While challenges remain-such as scaling scenario coverage, embedding continuous monitoring, and aligning with international standardization-the work presented here marks a significant step toward more harmonized and more universally encompassing assessment frameworks for future mobility.

5.2 Future work

Further development is required to move the proposed cybersecurity-informed safety assessment from a conceptual method to a viable, operational procedure. The current formulation of the approach offers theoretical value, particularly through its use of co-simulation to identify critical scenarios and assess the predictability of risks during validation testing. However, for this method to deliver actionable benefits, its foundations must be extended and empirically evaluated. This includes a systematic investigation of its capacity to identify and prioritize relevant attack-driven scenarios within representative traffic conditions. Such work is essential to establish its utility in independent assessments and to support adoption in practical settings.

The broader motivation for continued research remains strong. Future work will aim to further develop assessment methods that can be applied progressively throughout the product development lifecycle of ADS. This involves aligning technical assessment strategies with evolving guidance, such as that proposed by VMAD [29], and supporting both near-term and long-term goals through tailored technology solutions.

Key directions include the continued investigation of virtual validation as a means of accelerated testing, particularly in relation to scenario databases tailored to specific operational design domains. There is also a need to extend the scope of assessment to include human-machine interaction, ensuring that interactions between users and automated systems are evaluated in realistic and safety-relevant conditions [50].

Finally, a promising avenue of investigation is the transferability of the proposed multi-concern and scenario-based assessment approaches to other domains beyond the automotive sector. Investigating how these methods perform under different operational constraints, regulatory expectations, and technological architectures helps automation transfer and evaluate the generalisability of the assessment methods, informing refinement. Such cross-domain insights will enhance the robustness and flexibility of the integrated assessment framework.

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